

Elicitation

Introduction:

Elicitation is a task that helps the customer to define what is required. To complete the elicitation step we face many problems like problems of scope, problems of volatility and problems of understanding. However, this is not an easy task. To help overcome these problems, we have worked with the Eliciting requirements activity in an organized and systematic manner.

Eliciting Requirements:

Unlike inception where Q&A (Question and Answer) approach is used, elicitation makes use of a requirements elicitation format that combines the elements of problem solving, elaboration, negotiation, and specification. It requires the cooperation of a group of end-users and developers to elicit requirements. To elicit requirements we completed following four works.

1. Collaborative Requirements Gathering
2. Quality Function Deployment
3. Usage Scenarios
4. Elicitation work products

Collaborative Requirements Gathering:

Many different approaches to collaborative requirements gathering have been proposed. Each makes use of a slightly different scenario .We completed following steps to do it.

- The meetings were conducted with the assistant librarian (program) of the central library, University of Dhaka; the librarian was questioned about their requirements and expectations from the automated book circulation system.
- The librarian was asked about the problems she is facing with the current manual system.
- At last we selected our final requirement list from the meetings.

Quality Function Deployment:

Quality Function Deployment is a technique that translates the needs of the customer into technical requirements for software .It concentrates on maximizing customer satisfaction from the Software engineering process .With respect to our project the following requirements are identified by a QFD.

Normal Requirements:

Normal requirements consist of objectives and goals that are stated during the meeting with the customers. Normal requirements of our project are:-

1. Accessible via the Internet.
2. Allow any user to search for items.
3. Allow Librarians to check items for valid users.
4. Allow valid users to login and logout
5. Restrict access to functionality of the system based upon user roles
6. Allow valid users that log in to renew items, reserve items, and view the items
7. Allow Librarians to generate reports on the items in the system (e.g., all overdue items, all missing items.)
8. The application only needs to be installed and maintained on one computer.
9. Help feature to explain what they are looking for
10. A product reference manual describing how to install, setup, and run the application shall be provided.

Expected Requirements:

These requirements are implicit to the system and may be so fundamental that the customer does not explicitly state them .Their absence will be a cause for dissatisfaction.

1. Maintain a database of all items in the library's circulation.
2. The system shall enable the Administrator to change a user's type to any user type.
3. The system shall enable the Administrator to change user passwords.
4. The system shall enable the Administrator to configure the due date calculation for an item
5. The system shall enable the Librarian to change the number of items each user can check-out
6. The system shall enable the Administrator to configure the fine/item/day for an overdue item
7. The system shall enable any Librarian to give Librarian rights to other users.
8. The system shall allow the user to log in based upon an assigned login id and password.
9. The system shall automatically compute the due date for every item
- 10.The system shall automatically set the user status to "ABLE TO BORROW" and "UNABLE TO BORROW"
- 11.The system shall automatically send e-mail to the user when an item is overdue.
- 12.The system shall compute fines automatically for overdue items.

- 13.The system shall automatically set the Item Status to “AVAILABLE” ,
“UNAVAILABLE” ,”RENEWED”,”MISSING”
- 14.The system shall enable Librarians to add, edit or delete an item.
- 15.The user interface of the system shall be easy to use and shall make use
of drop-down boxes, radio buttons, and other selectable fields wherever
possible instead of fields that require the user to type in data

Exciting requirements:

These requirements are for features that go beyond the customer's expectations and prove to be very satisfying when present

1. The user interface should provide appropriate error messages for invalid input as well as tool-tips and online help
2. The user interface should follow standard web practices such that the web interface is consistent with typical internet applications.
3. Offer log in with mobile phone
4. The system’s configuration shall be documented and updated as changes to the system are made due to patches, new releases, etc.
5. Connect user account with facebook or other social media

Usage Scenarios:

At first a user authenticates in our system by creating an account. If a user already has an account then he/she will log in the system with his/her own password and username. Then our system will search the book that is requested by a user. If the book is not found, the system will exit. Otherwise system will check the availability of the book.

In case of Issue the available book:

- If the book is available the system will check the user database to confirm about the validation of the user. For an invalid user the system will exit and for valid user librarian will generate a call slip number manually for the user.
- Then our system will change the status of the book and user in our database.
- The system also generates a return date for the book.

In case of Issue the reserved book:

- If the user wants to issue a book which is already issued by a second user, the user is supposed to reserve the book and gets a reservation number on the basis of waiting list which is already in the reservation queue for that book.
- User is supposed to check his reservation number on the notice board database in our system and list is put up on the library notice board also by the librarian. Whenever they get the book free from the second user they give the book to the user who have priority in the reservation queue and second user is supposed to pay fine if he/she is returning the book after due date.
- User is supposed to issue the reserved book within three days from the day when his/her turns on the reservation list otherwise his reservation will be cancelled and he will not get the book (if there is waiting queue for that book).

- Now the user issues the book and the system automatically generates the return date (+7 days) and the user is required to return the book within the due date otherwise fine is imposed on him by the librarians.

Elicitation work product:

The output of the elicitation task can vary depending on size of the system or product to be built. Our elicitation work product includes:

- Make a statement of our requirements for automated book circulation system.
- Make a bounded statement of scope for our system.
- Make a list of customer, user and other stakeholder who participated in requirements elicitation.
- Set of usage scenarios.
- Description of the system's technical environment.